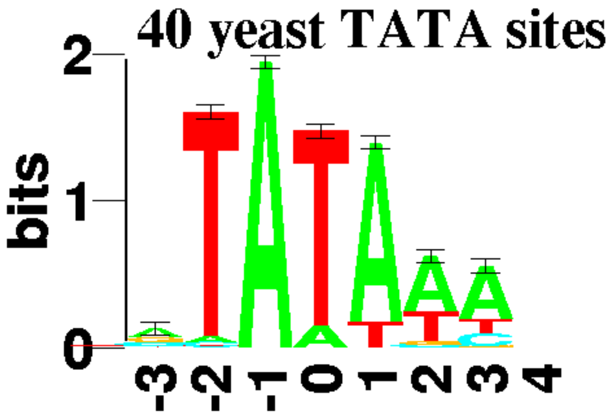
Multiple Sequence Alignment
MSA

What is MSA?

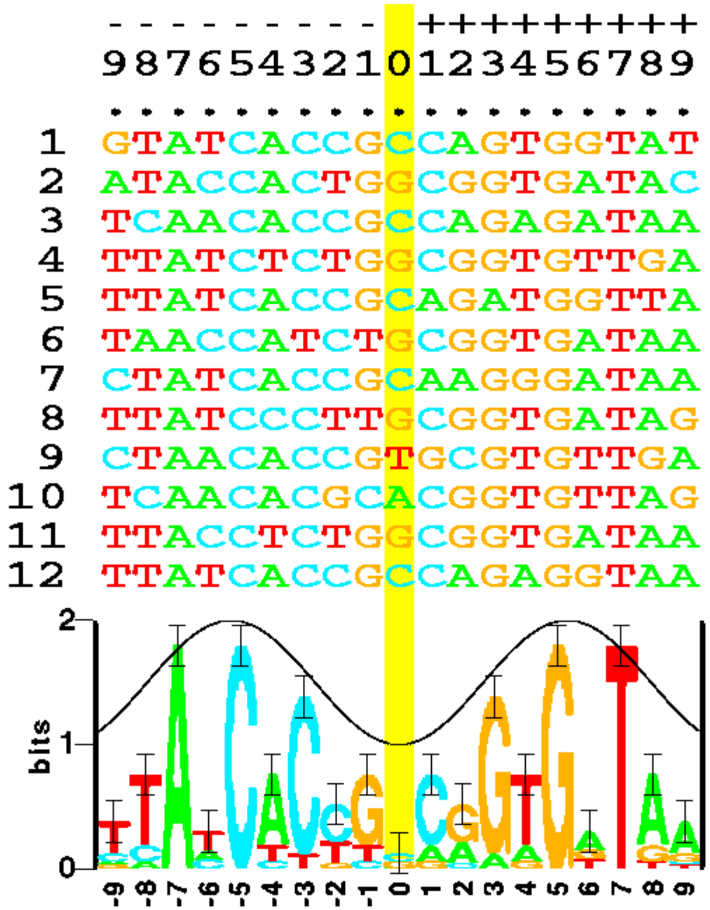
- Study more than two sequences at the same time
- MSA is more than a generalisation of pairwise sequence alignments
- Compare protein or DNA sequences for their evolutionary relationship
- Can be done as character-based study

Main use of MSA 1

- Detect common similarities in the sequences
  - Search for specific signals like
    - Promotor sequences, restriction enzyme sites,...
  - More sensitive than database searches
- Example:



Example for MSA of DNA sequence and resulting sequence logo:

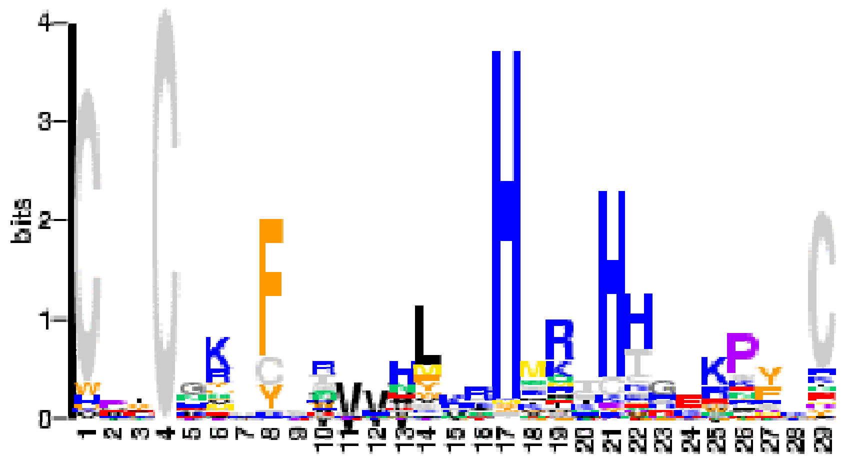


12 Lambda cI and cro binding sites

Fig. 1. Some aligned sequences and their sequence logo. At the top of the figure are listed the 12 DNA sequences from the P_L and P_R control regions in bacteriophage lambda. These are bound by both the cI and cro proteins [16]. Each even numbered sequence is the complement of the preceding odd numbered sequence. The sequence logo, described in detail in the text, is at the bottom of the figure. The cosine wave is positioned to indicate that a minor groove faces the center of each symmetrical protein. Data which support this assignment are given in reference [17].

Example for a protein sequence logo:

Cys-Cys-His-His profile: sequence logo form



**A sequence logo is a scaled position-specific a.a.distribution.
Scaling is by a measure of a position's information content.**

Main use of MSA 2

- Detect protein structure
  - alignments of proteins with known structure detect residues of functional importance

    => but only few structures are determined
  - use MSA of protein sequences to get structural information without knowledge of the structure
- Example:

  Homology-derived secondary structure of proteins (HSSP) database

Main use of MSA 3

- Detect dissimilarities which express mutations
  – differences between related sequences
  – can be used for phylogenetic analyses
- Phylogenetic analysis always starts with multiple alignments
  – more next in the next lectures

Task of multiple alignments

- Task: identify the corresponding regions of the given sequences
- example:

s_1 = `CGCTTTAATG`, s_2 = `ACGTTG` and s_3 = `GCTAGAG`

- There are many possible alignments:

```
CGCTTTAATG  |  CGCTTTAATG  |  CGCTTTAATG
ACGTTG      |      ACGTTG  |   ACGTTG
GCTAGAG     |     GCTAGAG  |   GCTAGAG
```

Multiple alignments

- Make them of equal length by introducing gaps
- Shift parts of the sequences against each other
- Similar regions become located above each other
- Quatify the quality of the alignments => use alignment scores
  - scoring systems
  - sum of pair scores
- multiple alignment of s_1 = `CGCTTTAATG`, s_2 = `ACGTTG` and s_3 = `GCTAGAG` would be:

```
CGCTTTAATG
-ACGTT---G
-GC-TAGA-G
```

MSA alignment scores

- Sum of pair scores
  - Score of MSA is the sum of scores for all pair wise alignments induced by the MSA
  - e.g score s_1 versus s_2 , s_1 versus s_3 , s_2 versus s_3 ,
  - Optimal alignment is the alignment with the maximum score
- scoring systems
  - Score DNA seq.: e.g.

    match +1, mismatch –1, gaps –3
  - Score protein seq.: e.g.

    Blosum62, PAM matrices, gaps

SP alignment problem

- Sum of pairs scoring leads to the SP alignment problem:
- Calculating the sum of pairs scoring
  - Grows exponentially with
    - Length of sequences and
    - Number of sequences to be aligned
- Solution: use heuristic approaches
  - Iterative alignments => ClustalW
  - Fragment based alignment => DIALIGN
  - Divide and conquer=> DCA

Iterative MSA (Clustal W) 1

Reduce problem to iterated application of pairwise alignment:

- Compute all pairwise alignments and their scores
- From the alignment scores compute a tree (for decision about the order of the pairwise alignment) = guide tree
- Align pairs of sequences by the brancing of the guide tree
- Stop once one large multiple alignment of all sequences remains

Iterative MSA (Clustal W) 2

- Advantages as compared to SP alignments:
  - Fast algorithm for many and long sequences
- Disadvantages as compared to SP alignments
  - Errors in early step cannot be eliminated later:
    „Once a gap always a gap“
  - No well-defined objective function for evaluation of this heuristic method
  - (possibly wrong) guide tree might bias the result of the alignment, which is critical for further phylogenetic analysis

Fragment based MSA (DIALIGN)

- Idea: first search for strong similarities between pairs of sequences
  - Done by computing local alignments
- Combine high scoring regions to colinear multiple alignments
- Again: (greedy) heuristic approach

Divide and Conquer Alignment (DCA)

- Idea: divide the original (long) sequence in a „divide and conquer” fashion
- This gives shorter and shorter sequences
  - that can be aligned with exact methods
  - results in sub-alignments
- Concatenate the sub-alignments to a complete alignment
- Works for ~ 15 sequences

Use of ClustalW

- Widely used for MSA of DNA and protein sequences

- Start for building evolutionary trees (inferring phylogeny)
  - Web server at EBI and Pasteur Institute
  - ClustalW easy to install under all platforms
  - ClustalX not so easy to install,
    - windows based version of ClustalW

- Visualisation of results

- Save results in proper file format (e. g. Phylip format)

Important web sites

- ClustalW www.ebi.ac.uk/clustalW
- DIALIGN http://bibiserv.techfak.uni-bielefeld.de/dialign/
- DCA http://bibiserv.techfak.uni-bielefeld.de/dca/
- Online multiple sequence alignments: www.pasteur.fr

MSA as a start for phylogenetic analysis

- proteins (DNA, RNA, organisms) are derived from common ancestors
  – and not “de novo”
- homologous sequences share common anchestry
- sequence similarity points to homology
  – although examples are known, where similarity is not based on homology e.g.

    RNAse and angiogenin

MSA for phylogenetic analysis 1:

- formats and tips for MSA
- input for MSA
  - Fasta format (save all sequences in one file)

  or

  - copy and paste from editor
  - use informative short headers as these later appear in resulting trees

MSA for phylogenetic analysis 2:

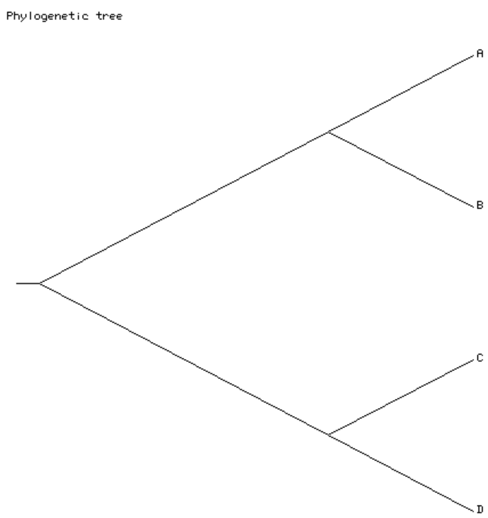
- Perform MSA and pay special attention to:
  - do not try to align sequences with sustainable length differences
  - cut off overhangs (nucleotides that are only present in one sequence)
  - pay attention to gaps, do they make sense
  - use visualisation tools like `jalview, clustalx` for quality check
  - save MSA in proper format (e.g. `phylip format`)

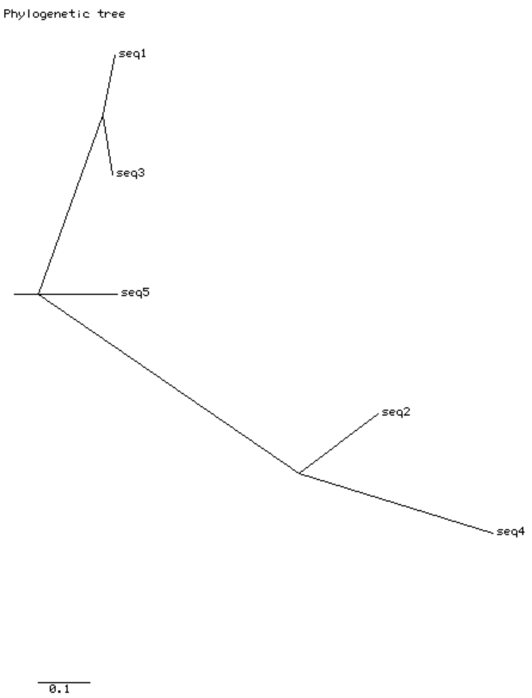
MSA for phylogentic analysis 2:

- Output (e.g. ClustalW)
  - scores seq 1 v. seq2, seq 1 v. seq3 …
  - guide tree and aligning scores (file `xxx.dnd`):
    - ((A:0.1,B:0.2)) or ((A,B),(C,D))
    - notation on next slide
    - use Phylodendron for visualisation
  - multiple alignment
    - file `xxx.aln`
    - store in phylip format:
  - phylip format:
    - 3 167 *# No of sequences and characters in dataset*
    - Humbeta *# 10 characters including blank spaces*
    - ----VHL-T?PREE ….. *# sequence, - = gap,*
      # ? = missing (unknown) character

Newick format of guide tree:

((A,B),(C,D))





(
(
seq1:0.01094,
seq3:0.00781)
:0.05932,
seq5:0.07193,
(
seq2:0.07204,
seq4:0.17634)
:0.23775);

Phylodendron: http://iubio.bio.indiana.edu/treeapp/treeprint-form.html